Calcurion Physics Worksheets

A-Level Physics

Topic: Fields

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. State Newton's law of gravitation. **[2]**

2. Calculate the gravitational force between two 5.0 kg masses 0.50 m apart. ($G = 6.67 \times 10^{-11}$) **[3]**

3. Define gravitational field strength. **[1]**

4. Calculate g at the Earth's surface. ($M_E = 6.0 \times 10^{24}$ kg, $R_E = 6.4 \times 10^{6}$ m) **[3]**

5. State Coulomb's law. **[2]**

6. Calculate the force between two charges of $+2.0\ \mu$C and $-3.0\ \mu$C separated by 0.10 m. ($k = 9.0 \times 10^{9}$) **[3]**

7. Define electric field strength. **[1]**

8. Calculate E at 0.20 m from a point charge of +5.0 nC. **[3]**

9. State the relationship between E and potential gradient. **[1]**

10. Define gravitational potential. **[1]**

11. Calculate the gravitational potential at the Earth's surface. **[3]**

12. Define capacitance. **[1]**

13. A capacitor stores 12 μC at 6.0 V. Calculate its capacitance. **[2]**

14. State the energy stored in a capacitor. **[1]**

15. Calculate the energy stored in a 100 μF capacitor charged to 12 V. **[2]**

Section B – Hard (Q16–25)

16. Derive $g = GM/r^2$ from Newton's law. **[2]**

17. A satellite orbits at height h above Earth. Write an expression for its orbital speed. **[3]**

18. Calculate the escape velocity from Earth's surface. **[3]**

19. Two parallel plates are 5.0 mm apart with p.d. 100 V. Calculate the electric field strength between them. **[2]**

20. An electron is released from rest at the negative plate. Calculate its speed on reaching the positive plate. ($e/m = 1.76 \times 10^{11}$ C kg^{-1}) **[4]**

21. Sketch the electric field pattern around an isolated positive point charge. **[2]**

22. Capacitors of 2 μF and 3 μF are in series. Calculate the total capacitance. **[2]**

23. The same capacitors in parallel. Calculate total capacitance. **[1]**

24. A 50 μF capacitor charged to 10 V is discharged through a 2.0 kΩ resistor. Calculate the time constant and the charge remaining after one time constant. **[4]**

25. Explain the similarity between gravitational and electric fields. **[3]**

End of Worksheet • *Show all working clearly.*